

Python Functions

Class Notes and Practice Questions

1 Introduction to Functions

A **function** is a block of reusable code that performs a specific task.

Real-life Analogy

- Employee Training → Function Definition
- Employee Actual Work → Function Call

2 Function Definition

```
def welcome_user():  
    print("Hello, user")  
    print("Welcome to the python world, let's begin")
```

3 Function Call

```
welcome_user()
```

4 Functions with Parameters

```
def welcome_user(name):  
    print("Hello", name)
```

5 Parameters vs Arguments

- Parameters: variables in function definition
- Arguments: values passed in function call
- Parameters = formal parameters
- Arguments = actual parameters

6 Multiple Parameters

```
def welcome_user(name, greeting):  
    print(greeting, name)
```

7 Default Parameters

```
def welcome_user(name, greeting="Hello"):  
    print(greeting, name)
```

8 Keyword Arguments

```
welcome_user(name="ram", greeting="Hi")
```

9 Multiple Default Parameters

```
def welcome_user(name, pro_lang="python", greeting="Hello"):  
    print(greeting, name)  
    print(f"Welcome to the {pro_lang} world")
```

10 Important Rules

- Default parameters must come after non-default parameters
- Required arguments cannot be skipped

- Keyword arguments can be in any order
- Positional arguments follow order

11 Practice Questions

Conceptual

1. What is a function?
2. Difference between function definition and function call.
3. What are parameters and arguments?
4. What is a default parameter?
5. What are keyword arguments?

Output Based

```
def greet(name):  
    print("Hello", name)  
  
greet("Ram")
```

Error Finding

```
def greet(msg="Hello", name):  
    print(msg, name)
```

Basic Coding Practice

1. Write a function that prints “Hello Python”.
2. Write a function to add two numbers.
3. Write a function to calculate square of a number.
4. Write a function with name and age as parameters.
5. Write a function with one default parameter.

Additional Practice

1. Function to Print a Line of Dashes

Write a function named `print_line()` that prints a horizontal line made of dashes (for example: `-----`). Call the function at least once to display the output.

2. Function to Print Your Name

Write a function `show_name()` that prints your own name when the function is called. Call the function to verify that it works correctly.

3. Function to Double a Number

Write a function `double_number(num)` that takes one parameter and prints the double of the number passed to it. Call the function with different values.

4. Function Using Positional Arguments

Write a function `add_numbers(a, b)` that takes two numbers as parameters and prints their sum. Call the function by passing arguments according to their position.

5. Function Using Keyword Arguments

Write a function `student_info(name, course)` that prints the student name and course. Call the function using keyword arguments so that the order of arguments does not matter.

6. Identifying Errors in Function Definitions

Study different function definitions and identify the errors in the following situations:

- A default parameter is written before a non-default parameter
- A function is called without passing required arguments
- More or fewer arguments are passed than required

Explain why each error occurs.